Maya Chen

Staff Software Engineer, Backend — Ledger & Payments Infrastructure

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KEY TECHNICAL IMPACT

- **Ledger Engineering:** Led the year-long rewrite of the double-entry ledger core at a 3.2M-account banking platform — 3,000 → 11,000 TPS sustained, p99 180 → 60 ms — proven equivalent over a nine-week parallel run.
- **Audit-Grade Accuracy:** Compared 2.8B journal entries across old and new ledgers with controller's-office sign-off on every amount-touching difference — zero unexplained mismatches through month-end and year-end close.
- **Operational Excellence:** Led the org-wide on-call overhaul — alert rules 340 → 92, pages ~900 → ~65/month, MTTR 47 → 19 minutes — with the customer-impact incident rate unchanged.

PROFESSIONAL EXPERIENCE

Bluepeak Systems — Senior Backend Engineer, Payments Platform — Feb 2023 – Present

- Wrote the design doc and capacity model that set the year-long ledger rewrite in motion — load tests put the legacy write path's ceiling at 3,000 TPS, 14 months out — then led the Go/gRPC rewrite to 11,000 TPS sustained, p99 180 ms → 60 ms.
- Designed the hot-account fix that made it possible — append-only internal postings with an async balance aggregator, group commit, and deadlock-free lock ordering — ending the row-lock queues behind 70% of month-end p99 latency.
- Rejected three simpler-looking designs in architecture review — ledger sharding, a distributed SQL engine, and Kafka as the system of record — writing the trigger that reopens sharding when growth demands it.
- Proved equivalence with a nine-week parallel run: 2.8B journal entries compared via a continuous Kafka diff (1.4B events/day, 240 topics), zero unexplained mismatches for the final 14 days across month-end and year-end close.
- Authored the change-management evidence package internal audit accepted for the ledger, written so a non-engineer could trace each mismatch from detection to sign-off; external auditors sampled 25 entries at year-end with no findings.
- Led the cutover from a 23-step runbook — 41-second write pause, no duplicate postings — then decommissioned the legacy Java service and its 24 VMs, holding a reverse-shadow rollback for three weeks.
- Tuned per-partition autovacuum after the first month-end on the new path spiked p99 to 95 ms; the next close held 61 ms.
- Traced a recurring reconciliation gap worth ~$1.2M/year in manual review to clock-skewed settlement events; fixed it with monotonic per-account sequence fences and documented it legibly for auditors.
- Led the last 31 VM-hosted services onto Kubernetes (~900 pods); capacity planning from measured p95s shrank the fleet 22%.
- Led the org-wide on-call overhaul: migrated ~40 CPU/disk rules to SLO burn-rate conditions, cut alert rules 340 → 92, rotations 7 → 4, runbooks CI-gated per pageable alert — pages ~900 → ~65/month, MTTR 47 → 19 minutes.
- Designed the backend hiring loop (now the company standard); interviewed ~90 candidates for 9 hires on the platform.

Northwind Labs — Senior Backend Engineer, Data Platform — Aug 2020 – Feb 2023

- Planned and drove the Kafka → Redpanda cutover of 96 topics and 12 consumer groups (peak 40K events/sec) through a six-week dual-write window: produce p95 28 → 11 ms, spend $214K → $131K/year on S3-tiered storage, zero data loss.
- Ran the three-month multi-region active-active Postgres evaluation and recommended against it with measured conflict data; shipped regional tenant placement with drill-verified automated failover instead — RTO 90 s, RPO 0 in-region, 74-second real failover.
- Cut end-to-end consumer delivery p95 3.8 s → 900 ms by resolving partition skew, unbounded polling, and consumer misconfiguration.
- Led a data workstream (5 → 8 engineers); mentored three engineers to promotion; wrote the org-wide promotion-doc template.

Cartway — Backend Engineer → Senior Backend Engineer, Payments — Oct 2017 – Aug 2020

- Owned the RCA of a $486K double-charge incident (1,912 transactions, all refunded within 40 hours) and led the six-month rebuild of the capture path on end-to-end idempotency keys in Postgres — zero duplicates in the 8 months after launch.
- Justified same-transaction Postgres idempotency over Redis and DynamoDB in a one-page decision doc — 4 ms slower at p99 for a correctness guarantee that cannot drift from the money.
- Built nightly payout reconciliation nine weeks from finance's request to production; unexplained balance days fell from ~6/month to zero and ~10 analyst-hours/week of manual spreadsheets were retired.
- Co-led carving a Ruby monolith into 14 Go services by transaction boundaries; ledger kept in partitioned Postgres (~4 TB).

Harborlight Software — Software Engineering Intern — Summer 2016

Freelance Web Development — ~20 sites for local nonprofits, part-time during school — 2014 – 2016

SKILLS

Languages & APIs: Go (primary since 2018), Ruby, gRPC

Data & Streaming: PostgreSQL at scale (partitioning, replication, failover), Kafka, Redpanda, S3 tiered storage

Infrastructure & Ops: Kubernetes (migration & operations), SLO burn-rate alerting, capacity planning

Financial Systems: double-entry ledgers, idempotency, reconciliation, parallel-run migrations, audit evidence

EDUCATION

BS, Computer Science — Alder Ridge University — 2013 – 2017

Systems coursework — databases, operating systems, networks; teaching assistant for the databases course in senior year.